

Benchmarking SATI.c

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System Specifications:

- OS Architecture :64-bit
- Processors: 12th Gen Intel(R) Core(TM) i5-1230U
- Number Of Cores: 10
- Number Of Logical processors: 12

Benchmarking: The program was compiled and executed using MS-MPI. For each processor count $p=1-5$, the program was run five times using “`mpiexec -n P MPI_Test.exe`”. The execution times were recorded and the average time for each processor count was calculated. All data, calculations, and graphs were created using Microsoft Excel.

- **Figure 1** shows the raw timing results for each run.
- **Figure 2** shows the average time measured and theoretical time.
- **Figure 3** shows the final graph comparing measured vs. theoretical performance.

Question #2		
P	T	Time (s)
1	1	0.044366
	2	0.042058
	3	0.043965
	4	0.049776
	5	0.045059
Tavg(1)		0.045045

Figure 1: Processor 1

mpiexec -n 2 MPI_Test.exe		
P	Tavg(P)	Theoretical(P)
1	45.044800	45.0448
2	27.590800	22.5224
3	32.594400	15.01493333
4	25.654400	11.2612
5	25.320400	9.00896

Figure 2: T_{avg} and Theoretical

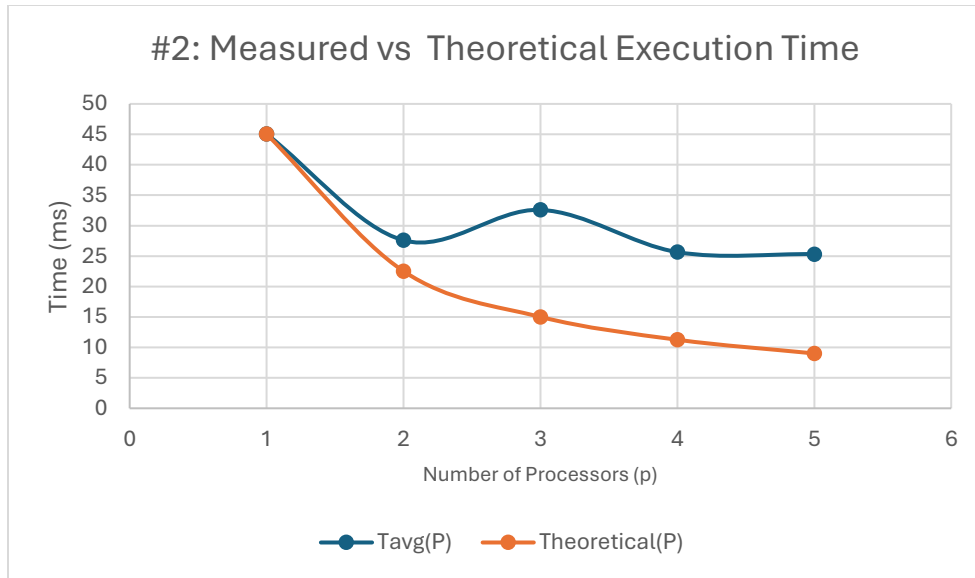


Figure 3 Measured vs Theoretical Execution

Summary:

In total, there are 527,787 six-digit combinations satisfy all constraints and can serve as valid identification numbers for the college. In addition, to evaluating the performance of the MPI program used to compute this value, I measured execution time using processor counts from 1 to 5 and ran it five times for each configuration. I also computed the theoretical execution time, where $T_{\text{theoretical}}(p) = T(1)/p$. Then plotting this data into a graph called measured and theoretical execution times, with the number of processors on the x-axis and execution time (ms) on the y-axis.

The theoretical curve decreases smoothly because it represents ideal parallel performance with perfect efficiency and no overhead. The measured curve also decreases as the processor count increases; however, it remains above the theoretical line and shows small fluctuations. Such as, the measured time drops at 2 processors, rises slightly at 3, then decreases again at 4 before stabilizing. This made the gap between the measured and theoretical curves noticeably larger at 3 processors, then settle into a more consistent distance for processors 4 and 5.

Interpretation:

The graph shows that execution time decreases as the number of processors increases, which is the expected behavior for a parallel MPI program. The fluctuations in the measured curve reflect normal variations caused by MPI communication overhead and operating-system scheduling. As more processors are added, overhead becomes a larger portion of total execution time, preventing the program from achieving ideal linear speedup.